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Innovation of cross-cultural music teaching methods based on artificial intelligence

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Abstract
This research is concerned with the innovative use of AI in cross-cultural music learning, more particularly in the creation of a multimodal learning model that combines audio identification, image analysis, and text understanding. The research is conducted with undergraduate music students from various ethnic backgrounds, and the teaching materials include a wide range of cultural music pieces. Sequence identification and semantic modeling are applied in the system to realize personalized recommendations and feedback interventions. As for model development, note recognition and cultural semantic decomposition are activities performed with a hybrid RNN-Transformer architecture. To perform a detailed analysis and to continuously adjust the performance, emotional expression, and cultural knowledge during the learning process, a teaching feedback mechanism and a cultural adaptation algorithm are implemented in the system architecture. The experimental data indicate that the learners' control of their rhythm, expressiveness of their emotions, and mastery of the culture have been facilitated to a considerable extent by the system's high recognition accuracy and the useful feedback offered. The experiment reveals that the cultural backgrounds of students play an important role in their learning, and hence teachers are given the opportunity to develop different teaching strategies according to the variations in culture. The developed system, as a technically robust model and application paradigm for cross-cultural music education, has shown great adaptability in the spheres of teaching logic, technical execution, and educational outcomes.

Keywords Artificial intelligence, Cross-cultural teaching, Music education

1 Introduction

Human cultural expression has music as its most important and basic aspect. It is the only one that can remain in the background but greatly help the intercultural dialogue and education events. Teaching music from different cultures has become a necessity rather than a side effect of musical skills development because it promotes understanding and integration among many cultures. On the other hand, the traditional methods of teaching are poor in content selection, instructional methodologies, and feedback mechanisms to meet the demands of cultural diversity in the globalized world. Students come to music that has different cultural backgrounds and are very eclectic in their tastes. That means teachers have to be more aware of the cultural contexts, be able to identify the stylistic elements, and, at the same time, understand the emotional expression through music in their teaching. The rapid development of AI technology has brought about new possibilities for cross-cultural music education through the use of multimodal data fusion, customized learning path creation, and smart recognition, among others. The involvement of AI has tremendously helped teachers recognize the cultural origins of their students, ascertain their individual learning styles, and generally stimulate flexibility and involvement in the classroom. Nevertheless, despite the entire package of theoretical and practical benefits, it is still advisable to deliberate deeply on the matter of AI's integration into classical music teaching across different cultures.

The interdisciplinary research has increasingly acknowledged the significance of the development of cross-cultural skills and the establishment of new artificial intelligence technologies. Zeba, emphasizing the utilization of AI in the manufacturing sector, opined that AI has literally boosted the manufacturing sector's ability to innovate and be more efficient through global data mining and the optimization of algorithms, thus leading the intelligent revolution of industries and providing a strong technical foundation for the associated research areas [1]. In his examination of the abrupt transformations caused by digital technology and AI in the public administration sector, Newman highlighted that AI is going to the extent of providing decision-making intelligence and automating processes in the public management systems, which are, in turn, increasing administrative efficiency and, thus, fostering organizational flexibility and responsiveness [2]. Analysis of Chinese business data performed by Zhai and Liu revealed that AI technology evolution has enormously raised the productivity of companies. Moreover, they pointed out that the extensive application of AI technology is a key factor in strengthening corporate competitive edges [3].

Schoen, in his research on intercultural communication and management, said that "standardized solutions do not apply to all cultural contexts," while he critically considered principled negotiations in cross-cultural situations. He proposed that communication and cooperation would be successful only if negotiators were well acquainted with the particular cultures and applied different negotiation methods that took into account the impact of cultural differences on the communication tactics during cross-cultural contacts [4]. To the realm of psychology and psychotherapy, from the Islamic point of view, Cucchi suggested a method that combined principles of Islamic faith with the cognitive-behavioral theory. He pointed out that in order to further the implementation of therapy and the incorporation of culture, it was imperative that the psychological interventions in cross-cultural contexts be aligned with the cultural values [5].In their systematic review on the future development of intercultural competence in higher education, Sabet and Chapman pointed out that, due to globalization, the development of intercultural competence has become a major goal in higher education, stressing the need for creative teaching and curriculum design that would not only enhance students' intercultural understanding but also their communication skills [6].

Vannuccini and Prytkova viewed the development trends of artificial intelligence technology from the technological systems perspective and suggested that the technology is evolving from an isolated one to an organized and integrated technology system. They also pointed out that the integration of different disciplines is a prerequisite for getting the most out of AI applications [7]. Li, relying on deep learning frameworks and affective teaching theory, developed emotional visualization techniques for music aesthetics teaching that not only improved the learning outcomes but also contributed to the development of the students' emotional cognitive skills, thus providing a practical case of merging intelligent technology with humanities education [8]. By investigating technological convergence in Chinese manufacturing companies, Ma and Wu found that the incorporation of artificial intelligence (AI) was a major factor contributing to this convergence, which subsequently reinforced the primary forces of corporate innovation ecosystems [9].To safeguard the reproductive rights of different communities, Homanen critiqued the application of AI in assisted reproductive technologies through the lens of reproductive justice, stressing that moral and social justice issues need to be considered thoroughly [10].

Researchers are currently investigating the huge impacts of AI technology innovations in various fields such as industries, management, education, and healthcare. They are also stressing the necessity of technology application and capability development in cross-cultural contexts. Hence, the study provides different angles and theoretical backing for future explorations. Cross-cultural music education is now under numerous difficulties, one of which is the teaching materials that are mostly influenced by the local culture, thus not giving a systematic introduction to other musical systems, leading to cultural bias and making it harder for the students to understand different genres of music. In addition, students are not able to comprehend other cultures through the existing teaching methods that are more or less uniform and do not adequately take into account the various cultural backgrounds and the different learning styles. It is a problem with the traditional ways that the assessment systems are designed that they cannot quantify the students' learning outcomes in the areas of awareness of and sensitivity to the different cultures, as well as appreciation of the styles, in a scientifically acceptable manner. The faculty basically does not possess the mentioned necessities of the first-world countries: technical competence, creatively developed courses, and awareness of the different cultures. The disparity in the distribution of academic resources, especially the lack of resources on lesser-known languages and minor cultural music, significantly limits the variety and depth of teaching to very low levels. In order to get over the limitations of the traditional teaching methods, it is a must to have a new teaching model that combines the elements of technological intelligence, cultural sensitivity, and individual responses.

This project is aimed at coming up with a global music education method that uses artificial intelligence to enhance the learning process, measurement accuracy, responsiveness to individuals, and cultural versatility. Its focus is mainly on three issues: the first one is how to get a wide range of teaching content through the use of AI in the extraction of features and recommending different musical cultural styles; the second one is how to have a dynamic feedback system that would allow the program to change teaching approaches according to the students' cultural backgrounds and learning paths during the teaching; and the third one is how to set up a quantitative model for teaching evaluation that would assess very well the students' musical performance and cultural understanding abilities. The aforementioned project will not only include the construction of deep learning models, the processing of various forms of data, and the creation of a decision-support system for the respective problems, but also the research of very practical and scalable teaching methods that can potentially bring about a paradigm shift in music education across cultures.

The integration of pictures, text, and other forms of data through the usage of technologies such as RNNs and deep neural network attention mechanisms is an effective way to learn the different aspects of students' musical performance. This method of teaching would allow for the mixing of the different elements in the auditory, visual, and cultural domains. A large-scale teaching database with examples from different cultural backgrounds and regions will bring data gathering and model training into the process and thereby ensure resilience and wide application of model training. An AI-assisted decision-making module will also be a part of the teaching system, which will provide dynamic feedback and the matching of teaching approaches to students' needs. The teacher evaluation model will integrate grading schemes for different aspects, such as the correctness of the performance, the expressiveness of emotions, and the understanding of culture, and apply weighted algorithms to calculate the overall performance of the students.

1.1 Related work

AI is revolutionizing the music teaching and learning area with its main characteristics, such as creativity, personalization, and inclusivity. The tools like AIVA, MuseNet, and NaadSadhana offer features like getting personalized feedback, being suggested learning paths, and getting help with practice and improvisation. AI, for instance, can make understanding an Indian raga or a Western harmony much easier for both the students and the teachers by breaking down the complexities involved in these areas. The chapter recounts the tale of AI in music from its very beginning, when it was restricted to extremely basic computer compositions of the 1950s, all the way to today, when it consists of deep learning systems that can compose, analyze, and even accompany live musicians. Moreover, AI gives rise to new genres and creative iterations through its limited intervention. There are also the issues of creativity, ownership, cost, accessibility, and the training of teachers that the technology will need to overcome before it can be widely accepted. Some of the solutions that have been put forward are the use of open-source tools and the establishment of nondiscriminatory AI systems. Aided by AI, the future sees the dawn of new technologies like virtual concert simulations, adaptive learning tools, etc.; therefore, the importance of finding the right balance between innovation and human creativity is once again emphasized. AI has been ubiquitous with technology that transforms the landscape in the domain of music education for all [11].

Producing musical audio straight from neural networks is a difficult task, not because it cannot be done, but rather because the musical structure is too complex to be modeled coherently along the entire time scale. Still, most music can also be characterized as having a very high degree of structure and being mostly made up of discrete note events performed on instruments. In this paper, we present how, if we take notes as an intermediate representation, then we will be able to train different models that can transcribe, compose, and synthesize audio waveforms with coherent musical structure that is distributed over the entire time scale of six orders of magnitude (~ 0.01 ms (8 kHz) to ~ 100 s). The breakthrough in the development of the models is made possible through our release of the MAESTRO (MIDI and Audio Edited for Synchronous Tracks and Organization) dataset, which contains more than 172 h of excellent piano performances recorded with a very close time alignment (~ 3 ms) for note labels and audio waveforms. The combination of networks and datasets offers an encouraging path toward developing new expressive and interpretable neural models of music [12].

The accompanying text explains a deep reinforcement learning-based algorithm for generating online accompaniment, which is capable of real-time interactive improvisation between a human and a machine. The process of music accompaniments in real-time requires the algorithm to give the machine's part of the music in a series and at that very moment to respond to human inputs, which is a different case from offline music generation. We view the problem as a reinforcement learning one where the generation agent acquires a strategy to make a musical note (action) based on the context already created (state). A well-functioning reward model is at the foundation of this algorithm. Rather than defining it with music compositional rules, we do so by learning from monophonic and polyphonic training data. The model takes into account the harmony of the machine note generated with the human and the machine's context. So the experiments reveal that the algorithm can echo the human part while producing a melodic, harmonic, and varied machine part. Preference tests reveal that the music pieces created by the new algorithm are of a higher quality compared to those created by the baseline method [13].

2 Materials and methods

2.1 Data gathering and sample choice

2.1.1 An explanation of the study participants and experimental environments

The research involves 120 undergraduate music students from different cultural backgrounds in four cultural regions: East Asia, Europe, North America, and Africa. The sample covers four language contexts: English, Chinese, French, and Spanish. Rigorous measures are taken to eliminate the influence of cultural factors, such as years of music education, language habits, and cultural area, under the guise of anonymity. Participants must have had systematic music education instruction within the last year and have some experience with basic instrumental performance or voice training in order to prevent individual variations in musical ability from creating structural biases in data analysis [14].

Besides the basic musical training criteria, the participants' backgrounds were closely scrutinized for demographic balance. Information about different aspects of their interaction with the cultures (for instance, how regularly they listen to non-native music or go to intercultural performances), their previous experiences with AI tools (like digital music platforms and intelligent tutoring systems), and their socioeconomic status (in terms of access to musical resources and family income) was collected. The pre-study questionnaires were the tool used to collect the information regarding these factors, and then the analysis was performed to check their impact on the learning results. The presence of these factors not only expands the potential use but also boosts the credibility of the research findings.

In this research, phased teaching interventions and follow-up assessments are put together, and a 12-week study period with two 90-minute teaching sessions per week is the duration of the research. All the instructional activities are done in one experimental setting with the standardized course materials, the teacher training programs, and technological equipment setups. The course material includes three cultural styles—classical,

Table 1 Statistical table of basic characteristics of research samples

Scope of numbering	Number of samples	Age range	The cultural circle to which it belongs	Language background	Years of music study	Specialization
S001-S030	Thirty people	Age 18–22	East Asia	Chinese	Four to six years	piano
S031-S060	30 people	Age 18–24	Europe	English	3–5 years	vocal music
S061-S090	30 people	Age 19–23	North America	English	5–7 years	The orchestra
S091-S120	Thirty people	Age 18–25	Africa	French/English	2–4 years	percussion music

Table 2 Descriptive statistics of sample demographics

Cultural region	*N*	Age range	Mean age	SD	Specialization
East Asia	30	18–22	20.1	1.2	Piano
Europe	30	18–24	21.0	1.5	Vocal
North America	30	19–23	20.5	1.3	Orchestra
Africa	30	18–25	21.2	1.6	Percussion

folk, and current music. For each style, modules covering performance practice, cultural background explanation, and emotional awareness are included [15]. At the end of each training phase, an evaluation is done, being staged and focusing on three areas: performance, cultural awareness, and emotional expression. All the experimental methods are documented properly and shot on video in order to facilitate data comparison. The films are stored in standard formats for further identification and analysis. As shown in Tables 1and 2 below.

2.1.2 Data collection tools and process design

Achieving high-precision, structured, and repeatable data collection, we have built an integrated data acquisition system that covers modules such as audio input, video recognition, text recording, and interactive feedback. The audio data collection uses a high-definition microphone array (Share MXA910) to capture performance details and vocal expressions; for the video part, we employ a dual-camera system with panoramic and partial views (Sony HDR-CX900 and Logitech C920 Pro) to collect facial expressions, postures, and actions, which are used for emotion recognition and action analysis. All audio and video signals are encoded synchronously and stored in a local database, then uploaded to the teaching platform server for backup [16].

The teaching software platform is a cross-platform system based on local deployment, integrating the Open Pose motion recognition framework and the Aubio music signal analysis tool. It analyzes learners' performance rhythm, pitch, dynamics, and movement changes in real time, automatically generating technical feedback reports. The cultural understanding and text input collection sections use a questionnaire system and a speech transcription module. All text data undergo bilingual annotation and sentiment dictionary classification for deep semantic analysis [17].

The whole process adopts digital segmented input, and the data is automatically summarized at the end of each teaching cycle, and abnormal values are manually checked to improve the level of data quality control (as shown in Table 3 below). The data collection operation process is uniformly trained by the research team to ensure consistent recording methods, standardized equipment debugging, and controllable environmental interference.

Table 3 Data collection content and tool matching table

Data type	Collection tools/systems	Data content	Output format
aural signal	Shure MXA910	Pitch, rhythm, strength, timbre	.wav/.flac
vision signal	Sony HDR-CX900,C920 Pro	Posture, facial expression, body movement	.mp4
Text input	Online questionnaire system	Cultural cognition, understanding, and feeling, self-presentation	.txt/.csv
Facial recognition data	Open Pose + Teaching platform	Key points of posture, trajectory of movement	JSON structured data
Sound analysis indicators	Au bio-open-source audio tool	Pitch changes, beat points, performance delays	Numerical table form

2.1.3 Teaching content and evaluation index setting

The design of the teaching content is based on a dual standard of cultural breadth and musical depth, selecting typical pieces from traditional Chinese music, European classical music, African folk rhythms, and Latin American popular music. The cultural misrepresentations and stylistic diversity of the content are emphasized. Each category of pieces is accompanied by background knowledge videos, emotional introduction materials, and technical practice manuals, forming an integrated teaching path of "listening-learning-performance-analysis." The teaching materials are sourced from international public copyright databases and have been adapted and standardized by our research team [18].

The evaluation system constructs three core dimensions: expressiveness, cultural understanding, and emotional communication, assigning different weights to each to comprehensively reflect the development level of students 'intercultural musical literacy. Pitch accuracy, rhythm, and performance precision are all included in the expressiveness dimension, which is evaluated through algorithms that extract audio features; the students' comprehension of the cultural context of the music is assessed through test and interview content; and emotional communication is realized through speech emotion analysis tools and facial expression recognition as well. A weighted linear model is integrated into the scoring system as shown in formula (1): as shown in formula (1):

$$S = \alpha \cdot P_m + \beta \cdot E_i + \gamma \cdot C_c \tag{1}$$

The default SS weights of 0.4, 0.3, and 0.3 can be altered depending on the specific learning objectives. The$P_m$letters correspond to the following: E_i is for Emotional Expression Index, C_c is for Cultural Understanding Ability Score, I is for Performance Score, and SS is for Overall Teaching Evaluation Score. $\alpha, \beta, \ \gamma$This method, by default, not only gives rise to high-quality annotated data ready for the next cycle of model training but also guarantees the quality, impartiality, and depth of the assessment system.

2.2 Model building

2.2.1 Principal analysis and AI model type selection

In detail, model selection was primarily driven by the ability of the models to work with different modalities, the depth of capturing the context, and the skill in sequence modeling, taking into consideration multidimensional tasks like audio identification, sentiment analysis, and cultural content interpretation. Traditional Convolutional Neural Networks (CNNs) have the inherent weaknesses of being unable to properly decipher the movement through time, the changing of the context, and the variation of the musical performance's rhythm; however, they excel in image processing and short-term feature extraction. The behavior of the musicians' performance can be mapped very well by Recurrent Neural Networks (RNNs) and their variations, Long Short-Term Memory (LSTMs), since they work with time-series data and thus can capture the note changes, rhythmic continuity, and sequencing of the emotional expression in the music over time [19].

The Transformer Model, along with its self-attention mechanism, has shown superior generalization abilities in multilingual context modeling and cultural text processing as far as long-term dependence issues and semantic relations between phrases are concerned. In terms of application scenarios, CNN is primarily used for motion and facial expression capture, RNN is fit for note detection and tempo analysis, and Transformer is applied for interpreting the meaning of multilingual cultural materials.

To make it even clearer, the RNN model is capable of keeping the time and pitch changes throughout the music, allowing it to play with the rhythm and pitch very well. The Transformer architecture, introduced by Vaswani et al. (2017), employs self-attention mechanisms that rapidly and efficiently take advantage of long-range dependencies and semantic ties in textual data from various languages and cultures. Attention-based sequence modeling was first recommended by Bahdanau et al. (2014), which allows for the theoretical underpinning of the combination of temporal learning and semantic interpretation. These aforementioned sources not only solidify but also validate choosing RNN and Transformer for temporal and cultural modeling tasks in this system, as shown in Table 4 below.

2.2.2 Rationale for combining RNN and transformer

Although transformer-based systems are very good at picking up long-distance dependencies and global semantic relations, they sometimes fail to catch the subtle rhythmic continuity that is very important in the musical performance data. The case is quite the opposite with Recurrent Neural Networks (RNNs), especially LSTM layers, which are still highly competent in capturing short-term sequential patterns and even timing nuances. Hence, the research incorporates RNN modules for accurate temporal learning and Transformer encoders for high-level semantic and cross-cultural contextual understanding. The combined architecture takes advantage of RNNs' sensitivity to rhythm and Transformers' ability to reason about context, thus attaining a balanced performance across rhythm perception, semantic comprehension, and adaptive feedback generation.

2.2.3 Model architecture and algorithm design

The note recognition model adopts a standard RNN structure, consisting of an input layer, a hidden layer, and an output layer. Sequences of mel-frequency spectrum features

Table 4 Contrast of usually cast-off AI representations

Types of models	Superiority	Limit	The location was applied in this study
CNN	Local perception, fewer parameters, and fast training	Lack of sequence modeling capability	Facial image recognition and pose estimation
RNN/LSTM	Time-dependent modeling is powerful	There is a gradient disappearance problem	Note sequence recognition and emotion tracking
Transformer	Long-distance dependence is highly processed	The parameters are complex, and the training cost is high	Feature extraction and classification of multilingual cultural texts

that represent the distribution of the audio spectrum over time during the student's performance are sent to the input layer. The hidden layer uses LSTM units to retain information from earlier notes and identify rhythmic patterns in complex playing behaviors such as glissando and legato. The output layer achieves note classification through the SoftMax function [20].

The input dimension for structural parameters is set to 128, reflecting the 128-dimensional Mel frequency features. The hidden layer dimension is 256, and a three-layer stacked LSTM structure is configured to enhance the model's ability to fit long-term sequences. The number of output categories corresponds one-to-one with the note categories in the training samples. The formula for the time-series notes recognition model based on RNN is as follows:

$$h_t = \sigma(W_{xh}x_t + W_{hh}h_{t-1} + b_h), \quad \hat{y}_t = \mathrm{softmax}(W_{hy}h_t + b_y) \tag{2}$$

x_tIs the input spectral vector at period$t$, $h_t$is the LSTM hidden state, $\hat{y}_t$Is the predicted probability distribution of the note category at this moment, where W and b are the weight and bias parameters.

2.2.4 Training strategy and optimization algorithm

The model training process is based on a supervised learning mechanism, using annotated audio-note pairs for parameter optimization. The training objective is to minimize the error between predicted values and actual values. It ensures the model's sensitivity to note boundaries and rhythm changes, with the loss function being cross-entropy, which is well-suited for multi-class classification problems. The optimization algorithm is based on the Adam adaptive gradient descent technique, and it makes use of an initial learning rate of 0.001, mini-batch processing with a batch size of 32, a training cycle of 50 epochs, and the early stopping mechanism to avoid overfitting. The cross-entropy loss function is the one used in the model training evaluation formula as follows:

$$\mathcal{L} = -\sum_{i=1}^{n}\sum_{j=1}^{C} y_{ij}\log(\hat{y}_{ij}) \tag{3}$$

The numerical factors that are taken into account are the sample count (n), the range of note categories (C), the definitive label for the i in the sample of the j category (I), and the projected probability (P) of the model. To guarantee y_{ij}that the note recognition task among different categories during the training phase was performed at the same level of performance, the validation set metrics of accuracy, recall, and F1 score are computed $\hat{y}_{ij}$.

2.2.5 A strategy for multi-modal application

In order to support the complex context of cross-cultural music teaching, a multimodal input system is created that combines text semantic vectors, image posture data, and audio features. The audio part uses Mel spectrograms and MFCC features to capture changes in rhythm, pitch, dynamics, and other parameters; image data extracts changes in facial expressions and movement characteristics to evaluate the quality of emotional expression; cultural text inputs come from cultural background introductions, student learning notes, and reflective texts, which are embedded using the BERT model [21].

The multi-modal feature fusion stage employs a strategy combining weighted splicing and attention mechanisms. It projects information from different modalities into a unified semantic space and inputs them in parallel into the Transformer structure to achieve collaborative information modeling. This method enhances the modeling capability of the correspondence between cultural semantics and musical behavior while avoiding modal redundancy. In other words, the RNN, CNN, and Transformer modules perform separately for emotional detection, rhythm finding, and semantic comprehension tasks, but still, their results are projected into a shared semantic space and combined using an attention-based weighted splicing technique. This makes it possible for the tasks to be done in a coordinated manner rather than one by one. Consequently, the system can detect the intricate connections between emotion, rhythm, and cultural context, which is the fusion output as shown in Table 5 below.

2.3 Model verification and system implementation

2.3.1 Architecture design of teaching interaction system

The teaching interaction system adopts a modular architecture design to achieve intelligent recognition, real-time evaluation, and dynamic feedback in cross-cultural music education. The system is primarily composed of five functional modules: data input, behavior recognition, learning path analysis, cultural context parsing, and feedback push, each corresponding to different AI technology tools. The data input module processes audio, video, and text signals, converting multi-source data into a unified format. The behavior recognition module uses a combined RNN and CNN modeling structure to extract students' performance trajectories, rhythm control, and expression changes [22]. The learning path analysis module combines historical data with current performance to construct individual learning profiles. The cultural context parsing module performs semantic evaluation and deconstruction of cultural materials through the application of natural language processing techniques. The feedback push module, drawing on system recognition results, integrates a teaching decision model and provides personalized recommendations for instructors and learners.

The module employs data flow protocols that are integrated for the purpose of exchanging and synchronizing information. The system comes equipped with multilingual adaptation interfaces that are suitable for various cultural educational contexts. The architecture supports bidirectional interaction, providing real-time guidance to learners and behavioral analysis reports along with course optimization suggestions to educators, enhancing the efficiency of teacher-student interactions in cross-cultural teaching. As shown in Table 6; Figs. 1 and 2 below.

Table 5 Is an example of a multi-modal input feature matrix

Modal type	Input dimensions	feature	Representation method
audio frequency	128 vi	Rhythm, pitch, dynamics	Mel-Spectrometer, MFCC
picture	68 vi	Facial key point coordinates, limb posture	Open Pose Landmark
text	768 vi	Describe the cultural context and reflect on the text	BERT embedded vectors

2.3.2 AI intervention mechanism in the teaching process

To enhance real-time response capabilities in cross-cultural music education, the system is equipped with an embedded intervention decision-making mechanism to dynamically adjust teaching strategies during student learning. The intervention mechanism is based on multi-modal data collected by the system, constructing a profile of the student's current learning status and calculating the intensity of interventions through weighted indicators from multiple dimensions. The intervention dimensions during the teaching process mainly include technical performance errors (such as unstable rhythm or pitch deviation), emotional expression deviations (such as inconsistencies between facial expressions and musical emotions), and cultural understanding gaps (such as semantic incoherence or partiality in text expression). Each dimension has a base score and fluctuation threshold, allowing the model to dynamically determine whether to trigger the corresponding intervention program [23].

Performance patterns, rhythm synchronization, and cultural background resources are some of the intervention methods for learning through deaf teachers. The system, however, regulates the intensity of the interventions to prevent unnecessary disruption of the learning patterns. After each intervention, the system will conduct a short-term evaluation of the results, record behavioral feedback to optimize subsequent decisions. The weighted decision formula for AI intervention factors is as follows:

$$D = \lambda_1 \cdot F_a + \lambda_2 \cdot R_t + \lambda_3 \cdot S_s \tag{4}$$

D represents the system's comprehensive intervention value, F_aIs the technical performance error factor, R_t Is the emotional expression deviation value, S_s is the cultural understanding score difference; $\lambda_1, \lambda_2, \lambda_3$ These are the weight coefficients of the three dimensions. According to the experimental design, the initial weights are set at 0.5, 0.3, and 0.2, respectively, and will be deceptively adjusted through the system learning process. The intervention mechanism can effectively enhance the system's ability to identify individual differences in actual teaching, improve the relevance and adaptability of teaching feedback, and provide real-time, precise, and saleable intelligent support for music education across different cultural backgrounds.

Table 6 Mapping relationship between system functional modules and AI technology

functional module	Corresponding AI technology	Input type	Output content
Data input module	Multi-modal prepossessing model	Audio, video, text	Standardized feature vector
Face recognition module	RNN, CNN, Open Pose motion recognition	Note sequence, posture image	Technical performance analysis results
Learning path analysis module	Deep user portrait modeling	Historical performance, feedback data	Individual learning preferences and problem identification
Cultural context analysis module	Semantic modeling and sentiment analysis of BERT	Study notes, cultural material texts	Cultural understanding depth, text emotional tendency
Feedback push module	Teaching intervention decision system	Comprehensive model analysis results	Personalized practice suggestions and strategy adjustments

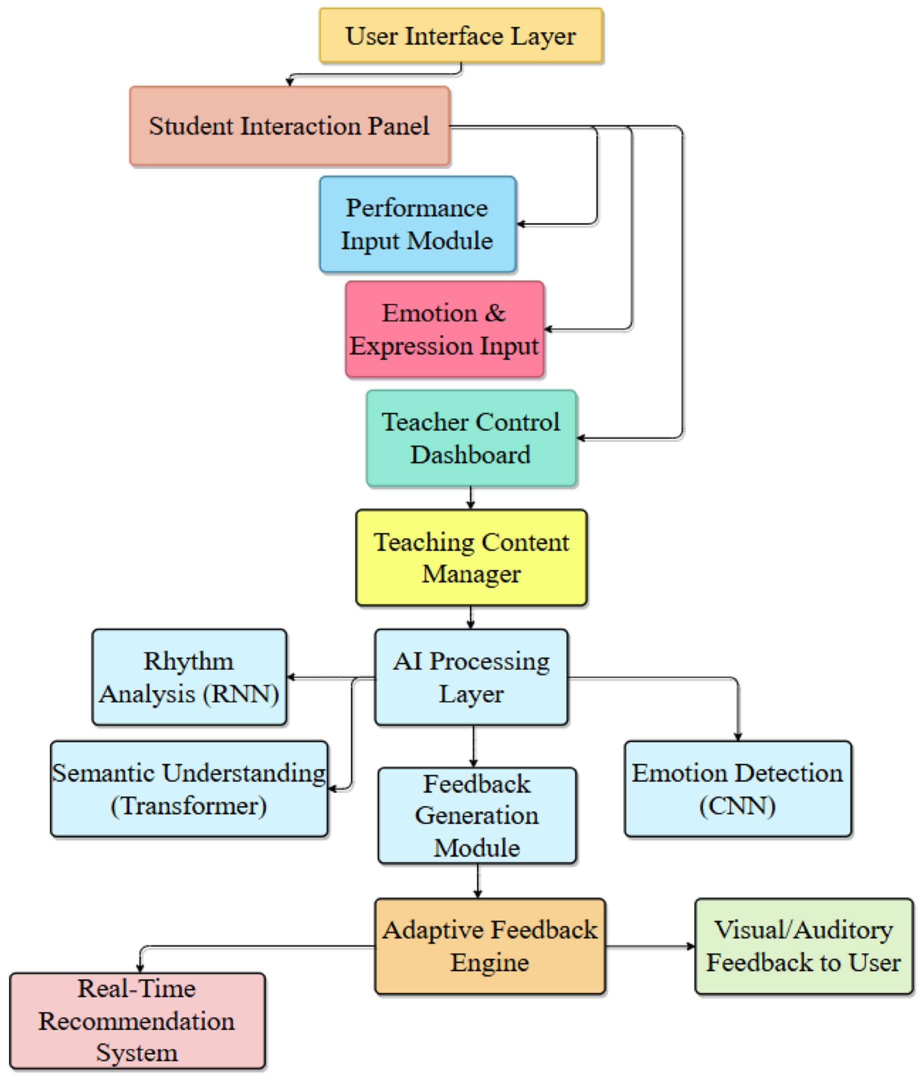

Fig. 1 Architecture of the system and flow of UI/UX of the teaching platform

2.4 Design of teaching application scheme

2.4.1 Intelligent push mechanism of multicultural music content

The effective implementation of multicultural music education relies on the extensive coverage and precise matching of content resources. In the teaching system constructed through research, the intelligent recommendation mechanism provides personalized content based on students' cultural background, language habits, and history of music learning. The input into the system operations is composed of the learners' selections, performance styles, cultural interest tags, and course performance. They get integrated alongside the metadata to form a multi-dimensional feature matrix. The back-end knowledge system is formed by cultural modules like traditional Chinese music, West African rhythm music, Latin American folk music, and European classical music. Various organized materials, including audio samples, research on cultural roots, score analysis, and performance clips, are being offered by each module as pedagogical tools.

The suggestions are made through a combination of collaborative and content-based filtering. The method not only determines how much people and things resemble each other by placing them into the overlapping categories of musical, emotional, and cultural

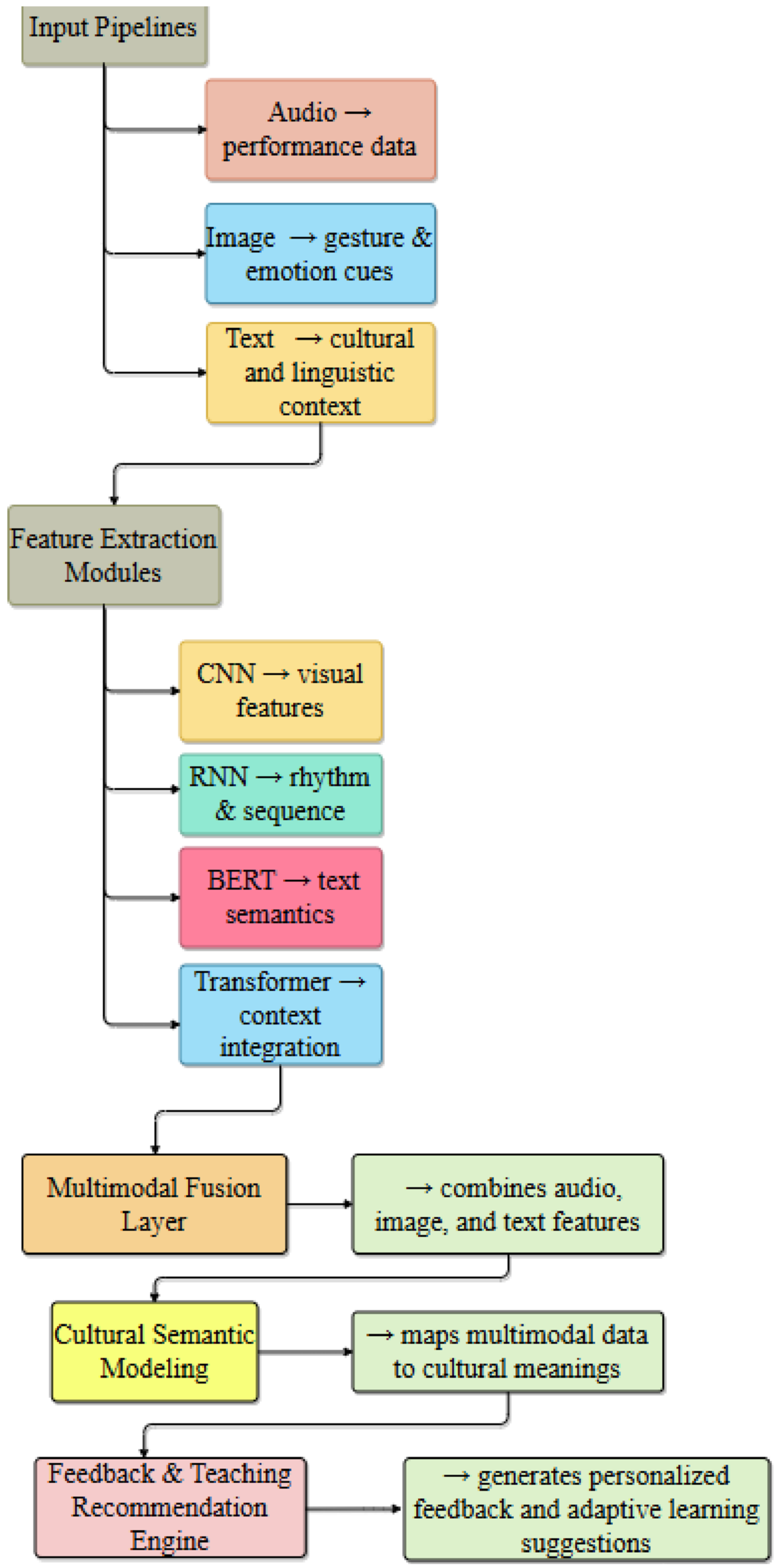

Fig. 2 Overall system architecture integrating RNN, CNN, BERT, and transformer modules

meanings. The next step is selecting the most similar songs which then proceed to the learning side. The system keeps the same learning load, but it can automatically detect the gradient of difficulty in the content and gradually suggest more difficult information. Through this method, students' cross-cultural aesthetic awareness, performing skills, and cultural range are not only enhanced but also diversified in terms of teaching materials which is a plus.

2.4.2 Interactive personalized feedback model

The feedback system is of utmost importance in music instruction. Teaching feedback in different cultures must be enriched with emotional expressiveness and cultural awareness alongside technical skills. This model is constructed on data regarding multimodal learning behavior, which has been systematically collected through real-time monitoring of textual expressions, performance, and facial emotions. The model's output is then transformed through a feedback engine into instructional recommendations, technical guidelines, or cultural explanations. It is subsequently presented in three different formats: text and picture reports, voice prompts, and video demonstrations.

The system personalizes feedback in a very unique way by adjusting the feedback frequency and structure according to student cognitive preferences, performance consistency, and recent training frequency. For students having difficulty in controlling the rhythm, the system provides synchronized practice suggestions based on a metronome. In terms of cultural understanding, feedback is given by providing the students with explanations of the cultural backgrounds, the similarities with the music of a similar style, which in turn leads to the students' better understanding. The feedback model has a dynamic updating mechanism that provides closed-loop optimization, swagger adjustments, and perpetual learning tracking. This system is effective in increasing engagement and motivation while at the same time giving accurate and efficient cross-cultural teaching support.

2.4.3 Teaching algorithms for the adaptation and control of cultural differences

The adoption of foreign ethnic music genres by students during real-world instruction, on the other hand, is very much a matter of individual differences and is characterized by differences in comprehension speed, imitative ability, and emotional engagement. The adaptive regulation algorithm, however, can provide a quantitative adjustment strategy to deal with such hurdles. The algorithm estimates the degree of adaptation of the students to the music style of the target culture based on their behavioral performance at that moment and their learning history beforehand. It keeps on adjusting its instruction, varying the rate of presentation, the range of the material covered, and the frequency of feedback.

Linguistic relevance, style acceptability, and perception of cultural distinctions are the three main factors that regulate the process. The setting of initial scores and movable thresholds is done for each dimension. The system not only calculates a complete adaptation coefficient but also modifies the dimension values according to the results of every teaching task in order to facilitate the subsequent delivery of teaching content and the reduction of intervention intensity. Thus, it is possible to achieve a balance between the cultural acceptance of the teaching methods and the learners' technical skills by using the instructional materials, which are very closely matched to the individual differences. The following is the formula of the model for the regulation of cross-cultural adaptation:

$$A_c = \mu \cdot (C_i - C_t)^2 + \nu \cdot T_f \quad (5)$$

$A_c$represents the current teacher's adaptation adjustment index, $C_i$the pupil's score in understanding the modern cultural style, the task's cultural complexity level, the emotional feedback frequency factor, and the adjustment coefficients C_t. The method here is to give an overall view of the individual.

teacher-student interaction and thereby promote the use of different teaching strategies personalized to the studentsμ, ν By the error-squared word and the emotional term dyads that reflect the challenge of cultural adaptation. The algorithm, by creating a dependable and proficient adaptive system for cross-cultural music education, enhances not only the effectiveness of the teaching but also the learners' cultural awareness breadth and depth due to the fact that it provides a technological grounding for diverse responses to different learners.

3 Results and discussion

3.1 Results

3.1.1 Examining the accuracy and convergence of model training

The note recognition model during the entire training process displayed very strong convergence and stability trends. The model is carrying out cross-validation of the training and validation data sets after every 50 rounds of training. After each round, the recall rate, accuracy, and loss value are recorded. The accuracy gradually increases from the initial value of 57.8% to 91.6% after the 40th round, with the subsequent changes becoming less significant; the loss value drops very quickly in the first 10 rounds, and after 20 rounds, it is already stabilized. The recall rate curve, just like the accuracy, indicates that the model has a very high coverage capability, along with the correct classification of the majority of the data. Each of the model's performance indicators suggests very slight changes only after the 45th cycle, thus meeting the convergence criteria. The early stopping technique for preventing overfitting is automatically engaged in the 48th round. Generally, the training outcomes prove to be very robust in distinguishing among different music segment types, and hence, the choice of the network architecture and the parameter settings is confirmed as suitable for the task. The gradient flow path seems to have been constructed correctly, as there were no significant occurrences of gradient explosion or gradient vanishing during training. The important signals from the training of the model are illustrated in Fig. 1 below (Fig. 3).

3.1.2 Comparison of performance differences in cross-cultural teaching

Students from different cultural backgrounds exhibit significant differences in cross-cultural music education. In total, there were four groups of thirty students each—one from East Asia, one from Europe, one from Africa, and one from Latin America—who participated in the study. The scores for three essential parts of the teacher assessment system—technical performance, emotional expressiveness, and cultural awareness—were compared. The finding was that the East Asian students turned out to be the best in cultural awareness, with an average score of 88.9 points, while the Europeans were better off technically, with an average score of 91.3 points. More precisely, the African

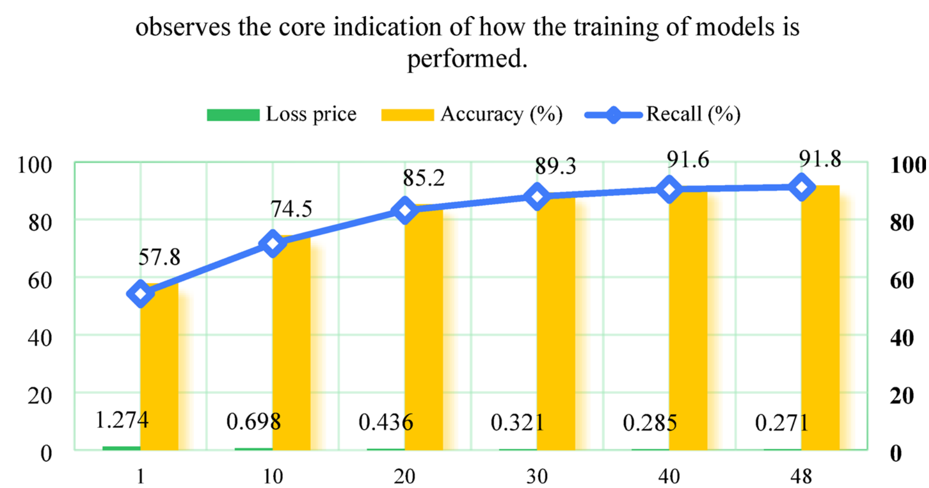

Fig. 3 Records the key indicators of the model training process

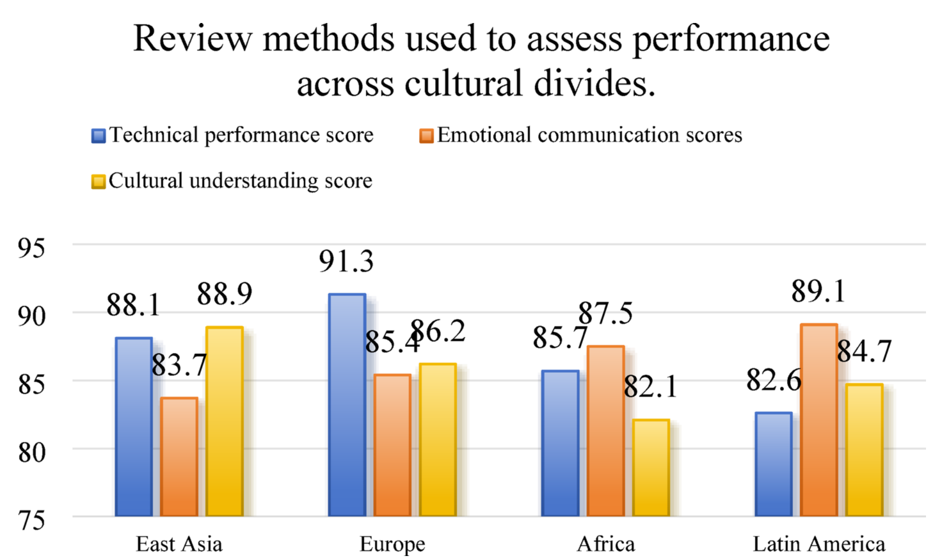

Fig. 4 Comparison of performance across cultural backgrounds on three assessment scores

group students exhibited a balanced performance in the skills-based courses, showing a strong edge in rhythm management and emotional expressiveness. Even though they were a bit less technically stable, the Latin American students were very much involved in emotional portrayal and cultural affinity. In a broader sense, the academic performance of students in particular musical styles is significantly influenced by their cultural backgrounds, and personalized, systematic adjustment methods are necessary for the teacher's overall effectiveness to be increased. The average scores of the four groups for each indicator dimension are illustrated in Fig. 4 below.

The performance of pupils from different ethnic backgrounds was compared, and the results were presented in Fig. 4.

3.1.3 Effectiveness evaluation of instructional feedback findings

The students are provided with suggestions for stage-wise modification of their performance, cultural awareness, and emotional expression through the instructional feedback system as part of a closed-loop optimization process. The evaluation adopts a one-test-feedback-post-test comparative model, with scoring and behavioral analysis conducted at the 3rd, 6th, and 9th weeks of the teaching cycle. Comparative data show that within two weeks after receiving system feedback, most students showed varying degrees of improvement across all three assessment dimensions. The most significant improvement was in rhythm accuracy, with a 12.3% increase; the correct rate of cultural knowledge questions improved by 9.7%; and the matching degree of emotional recognition and expression increased on average by 7.6%. The effectiveness of feedback did not differ significantly among different cultural groups, and the system model maintained stable output in multilingual and multicultural contexts, demonstrating strong adaptability and demonstrability. Figure 5 below presents the main indicator comparison statistics before and after the teaching feedback intervention.

3.1.4 System user experience rating statistics

The usability, interactivity, and cultural adaptability of the teaching system were evaluated using a standardized user experience scale (CUES-SF), with a total of 112 valid questionnaires collected. A five-point rating scale was employed to evaluate five survey dimensions: learning motivation, feedback response, cultural representation, simplicity of usage, and interface design. The results showed that the system earned high mean scores of 4.51 and 4.46 in the areas of "cultural expression accuracy" and "practicality of feedback guidance," respectively. The cultural training elements of the device were generally accepted by the users. Nevertheless, the "user-friendly interface design" rating was somewhat lower at 4.12, implying that some customers still do not need to change their way of operating. The feedback statistics in Figs. 6 and 7 provide the necessary data support for the future design of the system interface and user behavior interaction mechanism. This thus iteratively enhances the teaching experience and continues to encourage the improvement of platform functions.

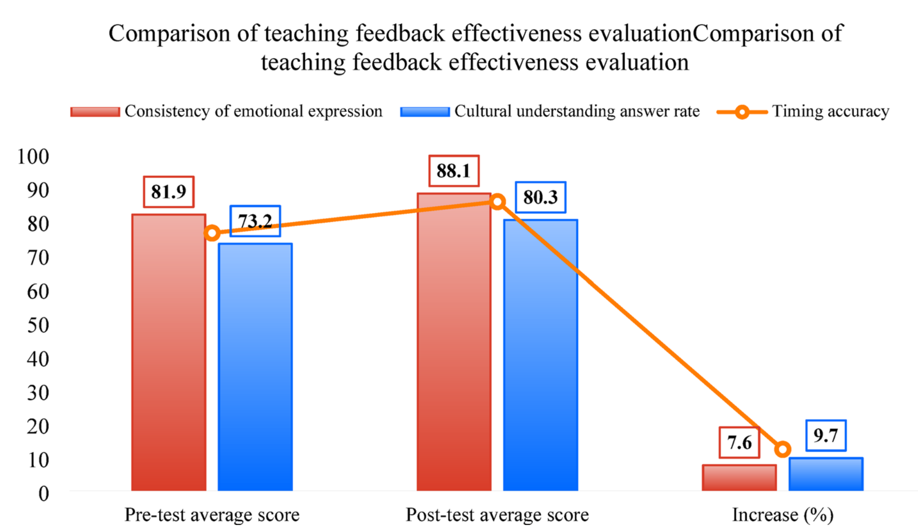

Fig. 5 Comparison of teaching feedback effectiveness evaluation

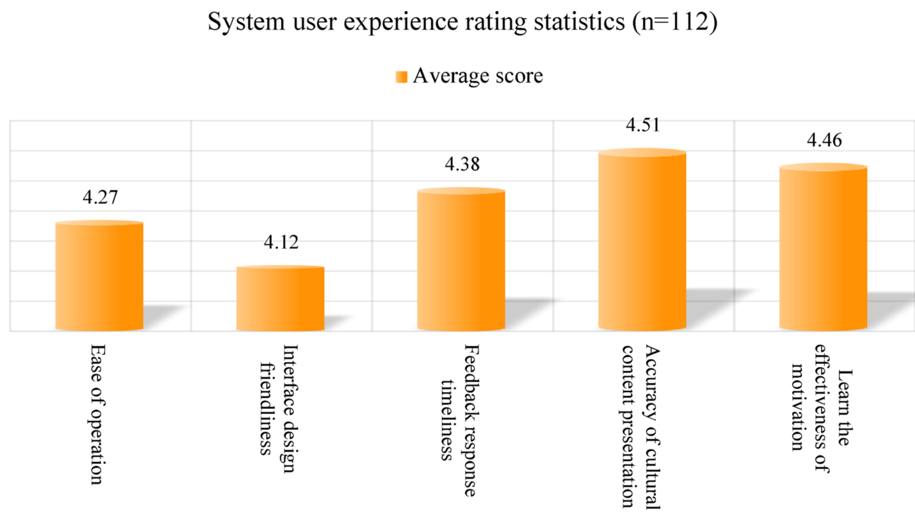

Fig. 6 User experience rating statistics of the teaching system across five evaluation dimensions—ease of use, interface design, feedback response, cultural representation, and learning motivation—based on 112 valid questionnaires using the CUES-SF scale, Fig. 4 shows the statistical results

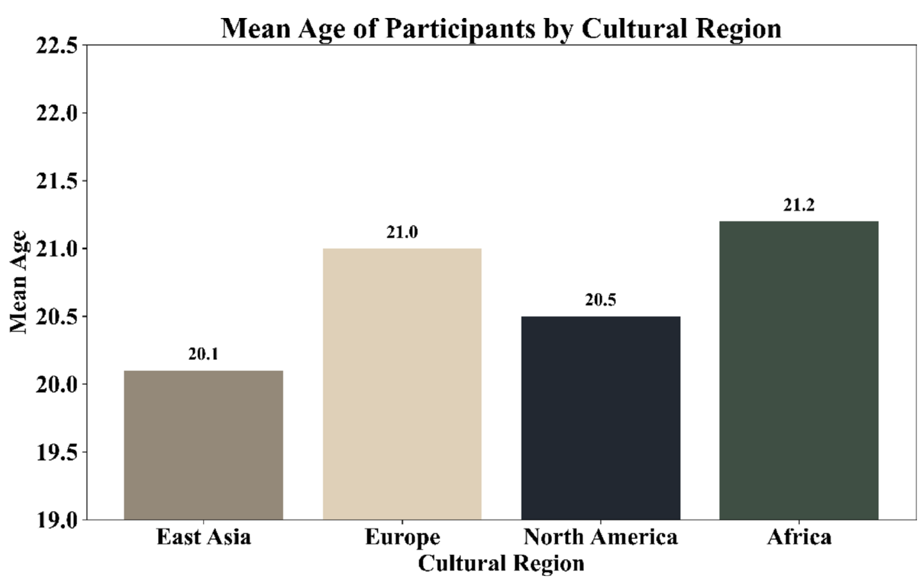

Fig. 7 Mean age of participants by cultural region

3.2 Discussion

3.2.1 Synopsis of study results and implications for instruction

According to the study findings, the AI-based cross-cultural music teaching model provides a variety of advantages, such as personalized feedback, enhanced cultural understanding, and increased effectiveness of teaching. The multi-modal data-driven performance recognition system can allow feedback from future interventions by capturing minute technical aspects of the students, such as rhythm, tone, and emotional expression. There are substantial variations in learning outcomes in teacher assessments due to cultural background variables. The students' different regional distribution of cultural acceptability, strategies of emotional expression, and performance pathways are a sign that the cultural adaptation processes of the system are quite effective in real-life scenarios.

The results previously discussed suggest that the main focus of the instructional insights is on three major areas. Firstly, the learner's background data should be wholly incorporated in the teaching methods to make a data-profile-based content distribution system. Secondly, the system should still be giving instant feedback and offering multi-dimensional reaction capacity to the learners to develop their self-awareness and adjust their behavior during the entire cross-cultural learning process. Thirdly, cultural presentation methods should no longer be teaching the previous information and should rather be going through music practice in the cultural settings to achieve the integration of technical learning and cultural experience [24].

3.2.2 Evaluation of research limitations and suggestions for improvement

Despite the aforementioned issues related to sample structure, universality of the model, and cultural dimensions, the findings of the research affirm the adoption and effectiveness of the system model. Mainly, there is an absence of experimental data, particularly from the stages of primary and secondary education, adult education groups, or non-music majors; and the samples are predominantly taken from higher education, with very narrow age and professional backgrounds. This situation constrains the model's usability in more widely applicable educational contexts. Furthermore, the range and complexity of the world's music areas are not properly represented in the training materials, which still largely restrict their focus to the works of a certain style with minimal cultural context.

In terms of model architecture, the note recognition and text semantic analysis modules are relatively independent, lacking true deep integration, which affects the performance of multimodal information synergy. The model architecture, though it consists of RNN, CNN, and Transformer blocks, still keeps the modular system design where the three components—emotion detection, rhythm analysis, and semantic understanding independently. The division of labor limits the system's ability to detect complex multimodal interactions. One research direction could be to develop a single deep learning system that integrates the training of all components via multimodal attention or end-to-end fusing techniques to enhance model consistency and interpretative power.

The precision of emotion modeling and cultural context analysis is still an area of research because the feedback system is still having difficulty managing complicated changes of expression and making cross-linguistic semantic analogies at a certain speed of reaction. The essential feature of the system concerning the feedback mechanism is that its architecture is still based on periodic interventions rather than on-the-spot, immediate changes that are dictated by student performance. This feature of the system greatly restricts its flexibility, and consequently, its ability to provide precise, instructor-aware assistance during live teaching sessions is greatly diminished. The active feedback algorithms' deployment would improve the quality of personalization and the timeliness of the teaching interventions to a greater extent [25].

The personalized paths of long-term learning are also influenced by the system's difficulty adjustment of teaching content, which relies on fixed label settings and does not have the capability of changing to learning rhythms in real-time. To do that, the model should not only have multi-modal attention processes but also extended sample types and cultural dimensions, plus an increased model capacity to analyze the relationship between cultural semantics and technical performance. In tricky cross-cultural environments, the feedback system's conversational performance and humanistic feel can be developed by adding APT-like language models to the mix, thus bolstering its context-generating skills in terms of culture.

4 Conclusion

This study systematically guides and empirically confirms the cross-cultural music teaching through the use of artificial intelligence. The study proves the coexistence of the multimodal recognition model for note sequence recognition, emotional expression analysis, and cultural text comprehension, which, in turn, significantly improves the accuracy of both the learners' responses during the teaching process and the scientific nature of the evaluation system. The results of the model's training show that the system has developed strong stability and flexibility in the complicated sequence modeling, the feature fusion, and the recognition of cultural differences. The technology indicates its ability to separate the performance traits of different cultural styles in numerous cultural sample comparison tests, thus providing students with appropriate teaching support according to their different backgrounds.

To enhance the efficiency of associating educational materials and the uniformity of learning experiences, the system design for educational applications applies a content delivery approach that is tailored to students' cultural inclinations and learning profiles to create resource pathways dynamically. The interactive feedback system with constant feedback of answers and suggestions for highly focused interventions helps the learners to pay attention to both the specifics of their performance and the cultural context. The algorithm for cultural adaptation modifies its parameters according to the cultural intricacy and the variations in performance so that it can efficiently solve cultural problems and reduce understanding time during the teaching process. Moreover, it provides the teacher in a multicultural environment with both academic and technical support.

Research models and teaching practices have shown great effectiveness, thus confirming the potential of artificial intelligence in cross-cultural educational settings. The backbone of the teaching system consists of the multi-modal integration, deep feedback, and dynamic adaptation, which together guarantee the dynamic synergy of the teaching content, process, and outcome. In the context of rapid globalization of education and cultural mobility, cross-cultural music teaching badly needs more adaptive technological tools. The study outcomes are highly systematic references for the preparation of course materials, the construction of educational platforms, and the adoption of modern teaching methods. Subsequent research should make the model's iteration capabilities more widely spread and tested in real classroom settings, scrutinize the dynamics of cultural production and smart teaching involvement in detail, and shape a new breed of education ecosystem that is both innovative and culturally sensitive.

Author contributions
Feng Liu is responsible for the framework creation, performance evaluation, result confirmation, and manuscript writing.

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Data availability

The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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